Data Management

01 Introduction & Course organization

Lucas Iacono. PhD

Graz University of Technology, Know Center Research, Austria

Institute of Human-Centred Computing

Slides credit: Dominik Kowald - Matthias Boehm

About Me



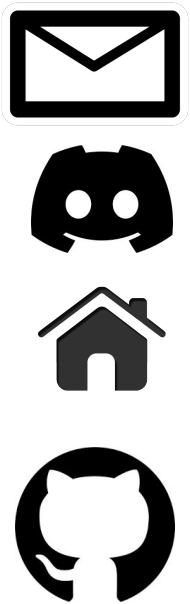




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How to contact me?

- liacono@tugraz.at
- lucasiacono
- SAL Building - 2nd Floor - Office 02 067
- https://lucasiacono.com



About Me

- **Degree in Electrical and Electronics Engineering** – University of Mendoza, Argentina (2007)
- **Doctor in Engineering (PhD)** - University of Mendoza, Argentina (2015)
- **Data Engineer** – Fiat Petronas PSG16 Race Team (2015 - 2016)
- **Associate Professor** -
  - Introduction to Technology – National University of Cuyo (2015 – 2019) - Argentina
  - Industrial Robotics – Computer Engineering – University of Mendoza (2008 – 2019) - Argentina
  - Postdoc – IoT Devices and UAVs applied to frost damage mitigation - Argentine Research Council (CONICET) - 2017
- **Know Center Research GmbH**
  - Senior Researcher - HAI Area (2019 - 2023)
  - Research Area Manager - Data Management for AI Area (2023 - Now)

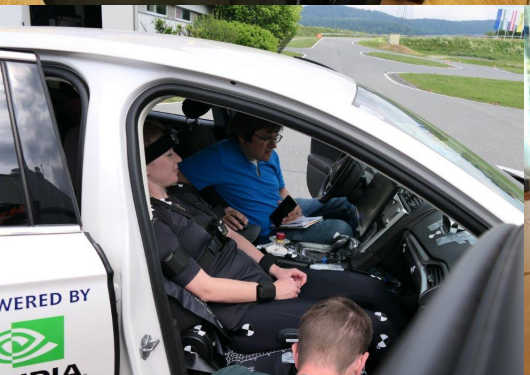

playseat


MOPAR
35
PETRONAS
Seguros
Federal

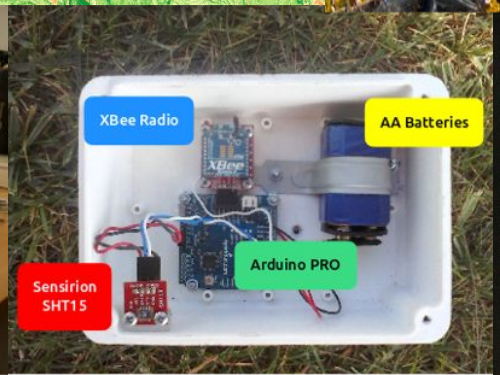


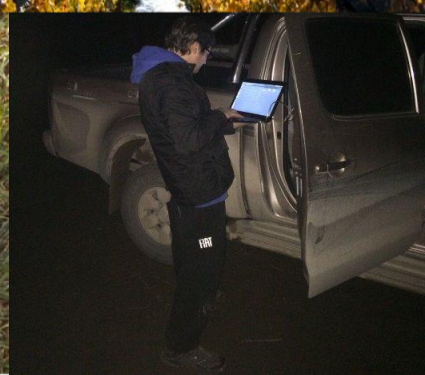





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XBee Radio
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SHT15



Data Management for AI @Know Center

Area Manager



Lucas Iacono

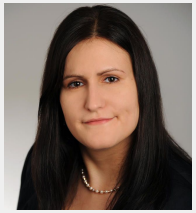
Senior Staff



Shafaq Siddiqi -



Mark Dokter -



Manuela Rauch

Data Scientists and Data Engineers







Alexander Hiebl - Lorenz Dirry - Santokh Singh



Saiful Islam -



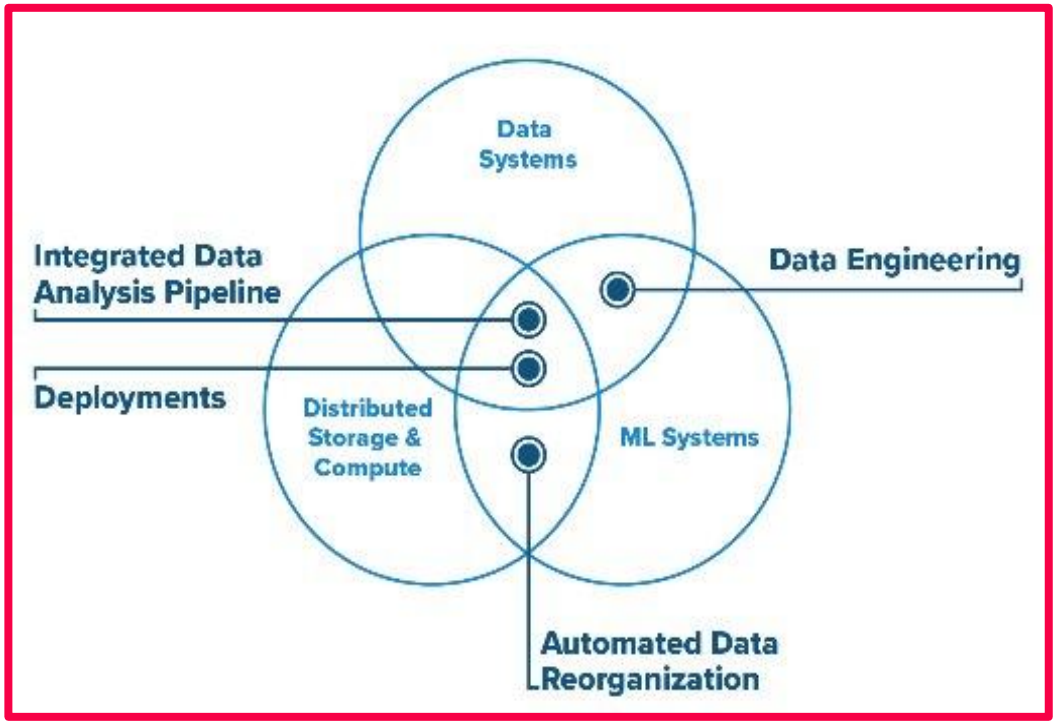
David Weissteiner -



Safiya Khanum

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Our research



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Agenda

- Short introduction to databases
- Course organization

Short introduction to databases

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History of data storage and management

- Essential for humans to store and retrieve information/data to make informed decisions
  - Should be as efficient as possible

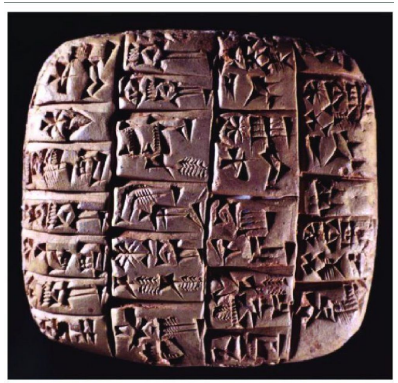
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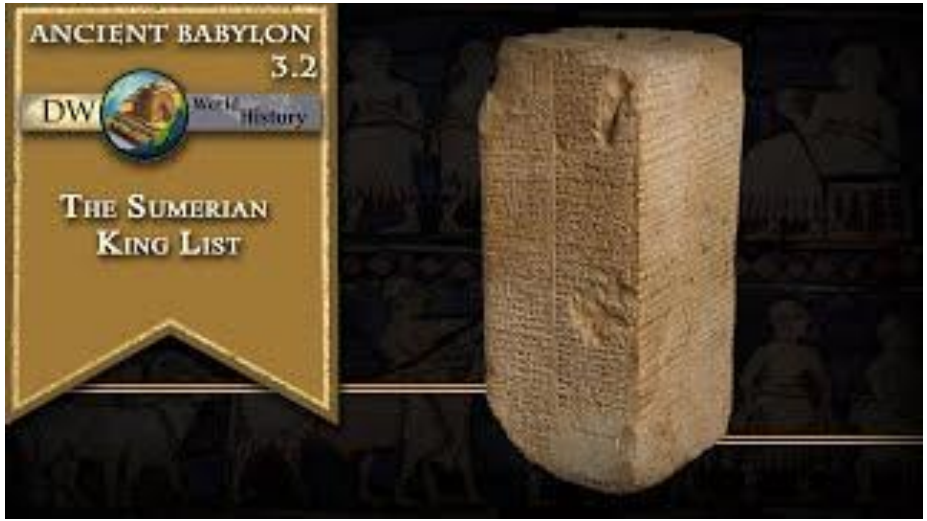
- Manual records of Sumerians are an early example of storing data (4000 BC using clay)



[archive.org]

History of data storage and management

- Sumerians: the king list

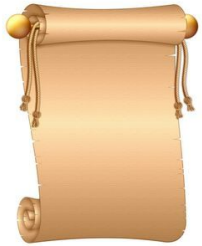


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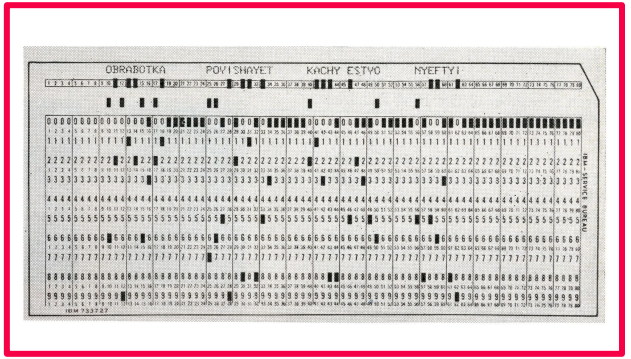
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- Data management in libraries → Dewey Decimal System
  - Developed around 1876 and still used today in 135 countries
  - Organizes books in libraries based on **subjects**
  - **3-digit numbers** for **main** classes (e.g., 300 is social sciences)
  - **Fractional decimals** for **detailed** classes (e.g., 323 is civil and political rights)



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- Punch cards (1900 – 1955)

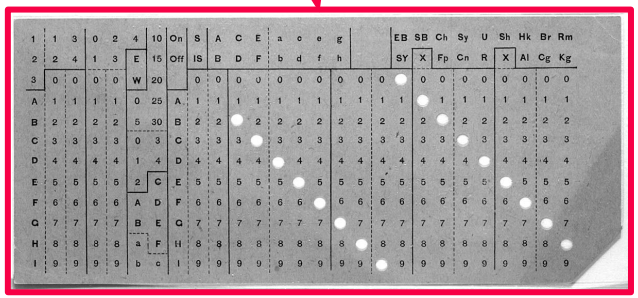


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- Punch cards (1900 – 1955)
- 1960 marks the start of computerized databases

Computerized databases

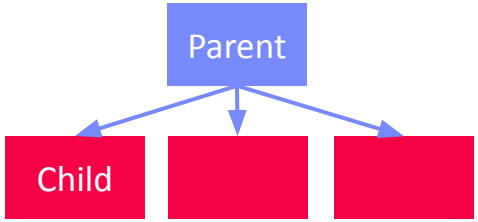
- Flat files
  - Flat → no connection with other files/records
  - E.g. Hollerit's punch cards database
  - Stored as sequential files → retrieval is slow
  - Example: .csv file



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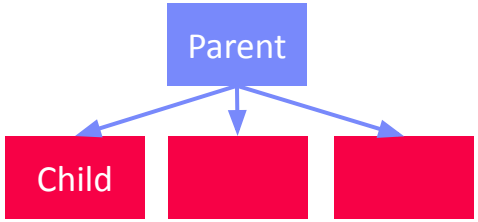
- Hierarchical model
  - Tree of records
  - E.g., IBM Information Management System
  - Each child can only have one parent



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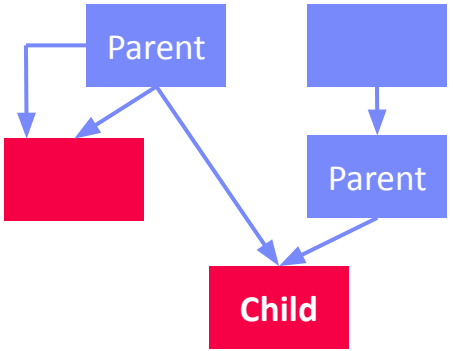
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  - **Issue:** Duplicates!



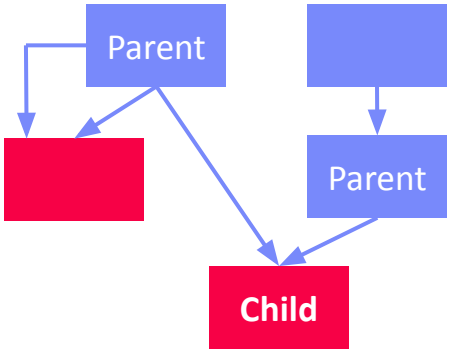
Computerized databases

- Network model
  - Graph of records
  - Developed by Charles Bachmann (Turing Award 1973)
  - Child can have multiple parents



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  - Developed by Charles Bachmann (Turing Award 1973)
  - Child can have multiple parents
  - **Issue:** Complex array of pointers leads to complex implementations



Relational databases (1970/80s – now)

School Table

ID	Name
S001	University of Technology
S002	University of Applied Science

Student Table

School ID	ID	Name	DOB
S001	UT-1000	Tommy	05/06/1995
S001	UT-1000	Better	16/04/1995
S002	UAS-1000	Linda	02/09/1995
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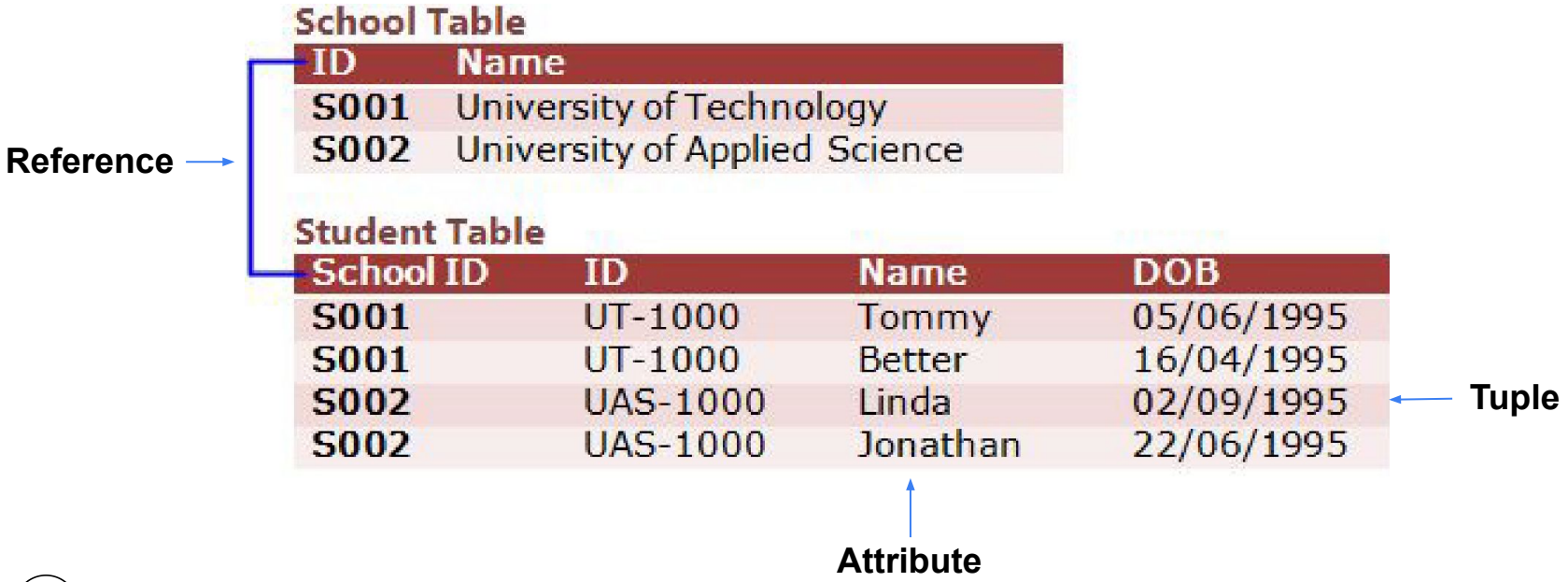
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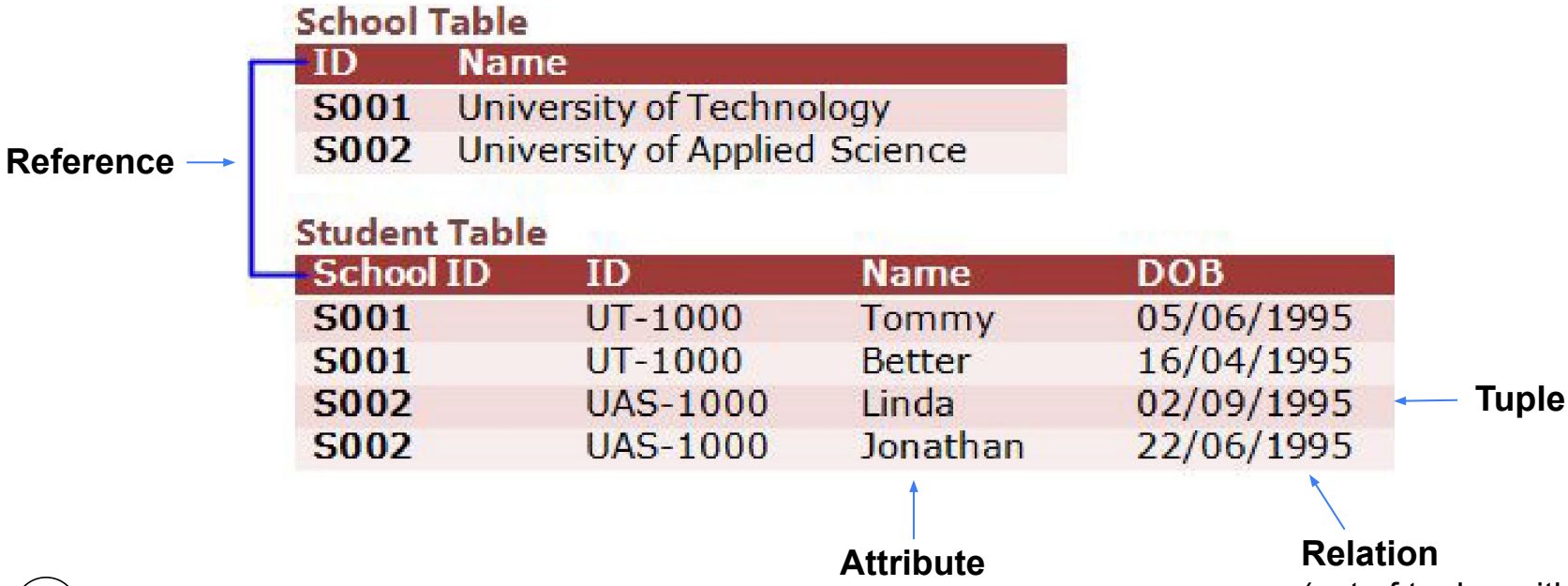
Tuple

Attribute

Relational databases (1970/80s – now)



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- Organize a body of data into simple tables of related information
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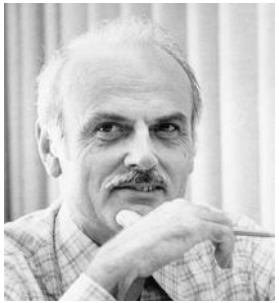
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Edgar F. “Ted” Codd @ IBM Research (Turing Award ‘81)

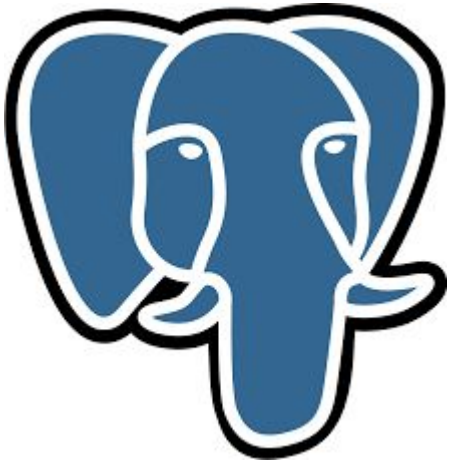


[E. F. Codd: A Relational Model of Data for Large Shared Data Banks. Comm. ACM 13(6), **1970**]

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Our relational database: PostgreSQL

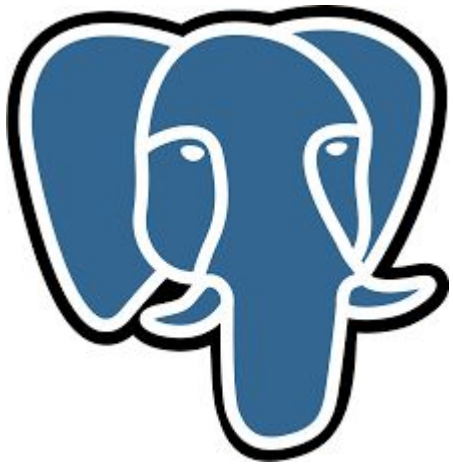
- Open-source and free relational database management system
- Easy to install, e.g., (https://www.postgresql.org/download/)
  - **See also Tutorial pdf in TeachCenter**



[postgresql.org]

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- Open-source and free relational database management system
- Easy to install, e.g., (https://www.postgresql.org/download/)
  - See also Tutorial pdf in TeachCenter
- Nice GUI (pgAdmin)
- Good support and user community
- More on this also in the next lectures



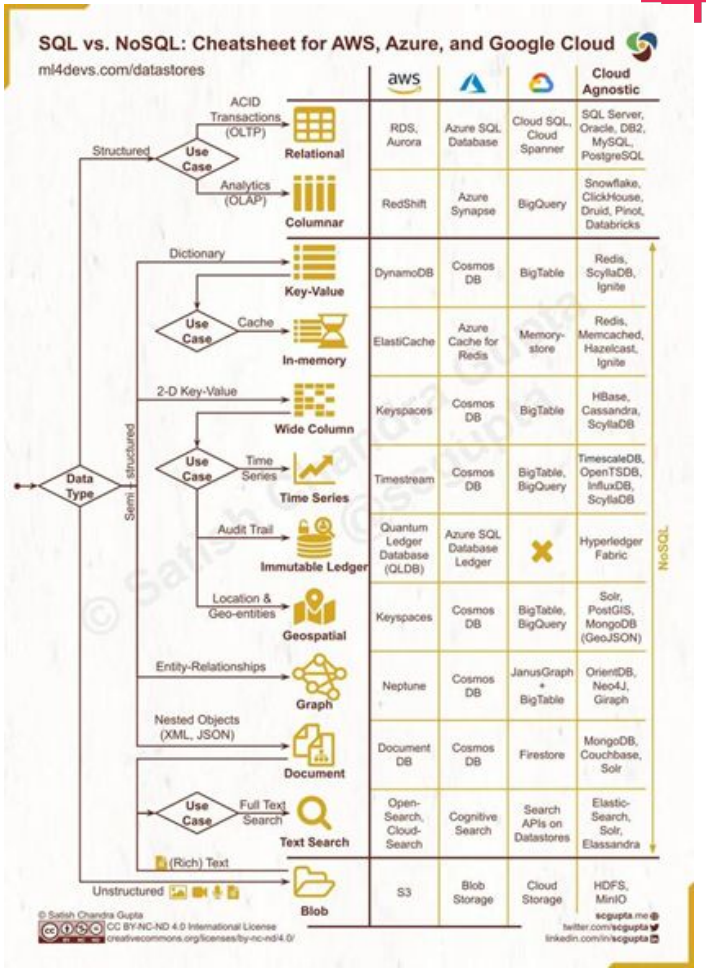
[postgresql.org]

Beyond relational databases (NoSQL)

- Data became bigger (BigData) and in more variety
  - Graphs (nodes/edges/attributes)
  - Time-series (sequences of observations)
  - Key-value stores (simple put/get/delete operations --» high scalability)
  - RDF stores (subject/predicate/object triples --» semantic Web)
  - Document stores (store and search in documents, e.g., Apache Solr)
  - Geo-spatial databases (store and retrieve geo-locations)
- NoSQL

Beyond relational databases (NoSQL)

[ml4devs.com]

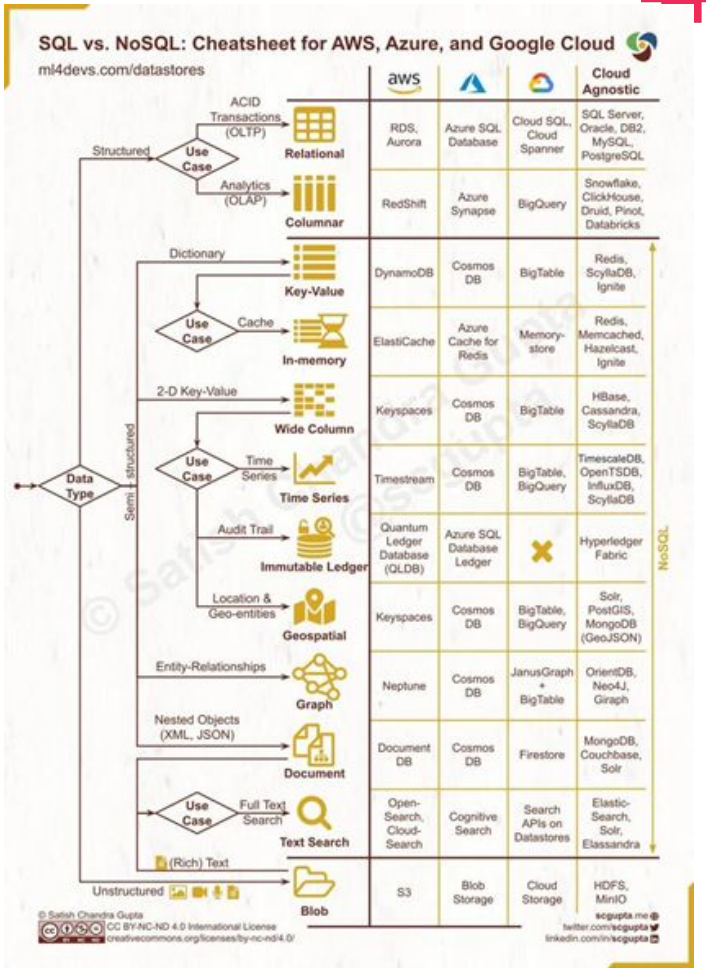


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Beyond relational databases (NoSQL)

- Our course: relational (PostgreSQL) and a bit of NoSQL
- Excurse to other types in Data Integration and Large-Scale Analysis courses

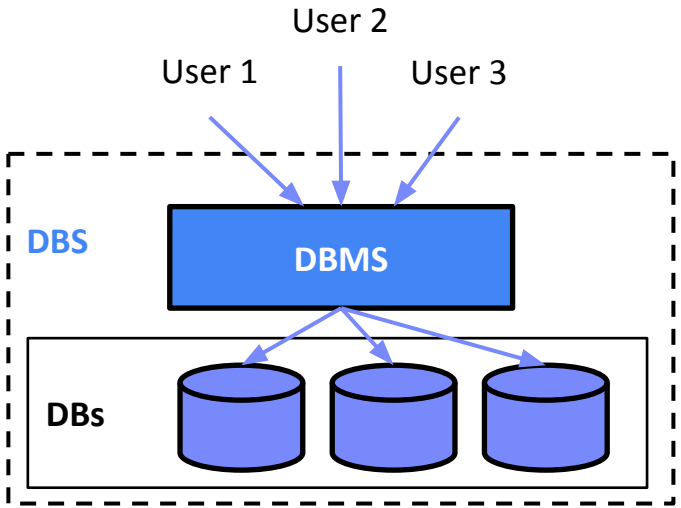
[ml4devs.com]



What is a database?

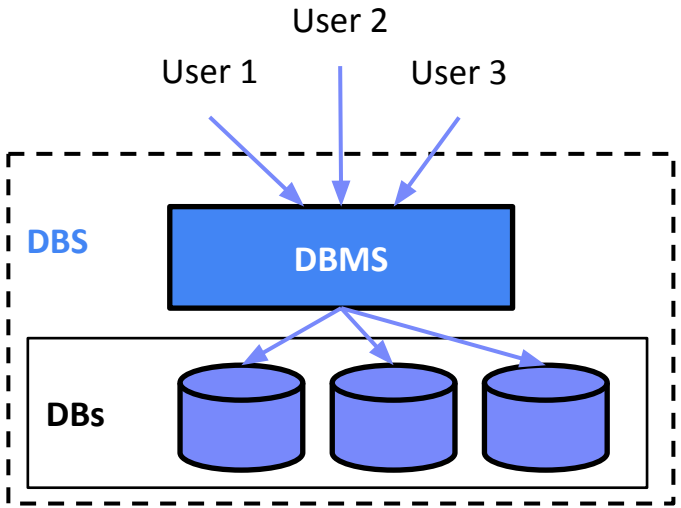
Definition of a database

- Database system (DBS): DBMS + DBs



Definition of a database

- Database system (DBS): DBMS + DBs
- DBMS: Database Management System (Software to handle DBs)
- DBs: Database (data/metadata collection to describe something)
- Note: DB also a short for DBS/DBMS



Why it is important?

- Important for data analytics / research

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  - Market Volume: **10-100B U$S**
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- Technical
  - Consistency and data integrity (no redundancy)
  - Performance and scalability
  - Multi-user operations and access control
  - Application development and maintenance costs

Readings (optional)

- Raghu Ramakrishnan, Johannes Gehrke: Database Management Systems (3. ed.). McGraw-Hill 2003, ISBN 978-0-07-115110-8, pp. I-XXXII, 1-1065
- Jeffrey D. Ullman, Jennifer Widom: A first course in database systems (2. ed.). Prentice Hall 2002, ISBN 978-0-13-035300-9, pp. I-XVI, 1-511
- Ramez Elmasri, Shamkant B. Navathe: Fundamentals of Database Systems, 3rd Edition. Addison-Wesley-Longman 2000, ISBN 978-0-8053-1755-8, pp. I-XXVII, 1-955
- Alfons Kemper, André Eickler: Datenbanksysteme - Eine Einführung, 10. Auflage. De Gruyter Studium, de Gruyter Oldenbourg 2015, ISBN 978-3-11-044375-2, pp. 1-879

Course organization

General information

- Title: Data Management
- Type: VO + KU (Lecture + Practical separated)
- Semester hours: 2 (3 ECTS) for VO, 1 (1 ECTS) for KU
- Offered in: SS

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- Studies:
  - Software Engineering and Management (bachelor)
  - Computer Science (bachelor)
  - Information and Computer Engineering (bachelor)
  - Data Science (master)
  - Others?

General information

- Teach Center & Discord:
  - Discord for questions, discussions: https://discord.gg/hHFPNCsdp8
  - Teach Center for material: https://tc.tugraz.at/main/course/view.php?id=1998
  - **E-mails only for important organizational questions!**
- Language of Instruction/Communication: English
  - Informal language (first name is fine)
  - Please use English when asking questions
  - Online availability: **Mo-Fri 9:00 to 17:00**
  - Hello and Thanks when asking questions are welcome!

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- Teaching and Learning Method:
  - Topics of the VO are complemented by **practical exercises using PostgreSQL (KU)**
- Objectives:
  - With successful completion of this course, students have gained a **basic understanding of databases, as well as data management processes**.

Content

- This course covers, primarily from a user perspective, foundations of **databases and data management** in terms of:

Content

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  - Overview of databases and data storage systems

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  - Physical design and tuning
  - Query and Transaction Processing
  - NoSQL databases and Data spaces

Course team

- Lecturer: Lucas Iacono



- Advisor Lecturer: Dominik Kowald



- Exercises (KU): Shafaq Siddiqi



- Student assistants (tutors):
  - Margarita Dubina
  - Toni Guli
  - Ema Srna







Assessment (VO)

- **Written exam (3 per semester) - dates:**
  - Exam 1: 23.03.2026 (i13)
  - Exam 2: 04.05.2026 (i13)
  - Exam 3: 15.06.2026 (i13 + other room if needed)
  - 3 more dates in **WS** (probably October, November, December)

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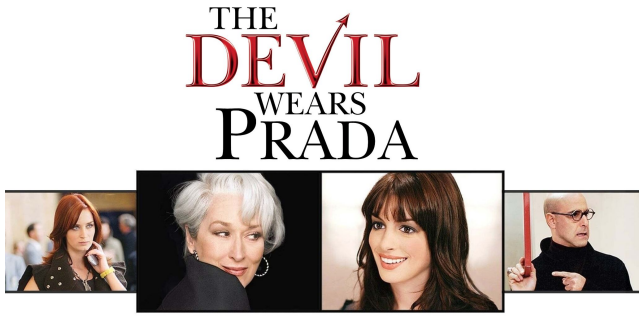
  - Not mandatory to attend lectures (but recommended)
  - Lecture videos available: https://tube.tugraz.at/paella/ui/browse.html?series=db3321ae-3049-46eb-9e8b-a452c761531a

Assessment (KU)

- 3 exercises (80 points) **--> individual**
  - Exercise 1: Data modeling (25 points)
  - Exercise 2: Data ingestion and queries (30 points)
  - Exercise 3: APIs and Tuning (25 points)

Exercises (KU)

- Dataset: **Multitable Ecommerce European Fashion**
  - Dataset from Kaggle
  - Simulates the operation of a fictitious fashion store operating in Europe via E-commerce and Mobile App.
  - Artificial and created for educational purposes.
  - https://www.kaggle.com/datasets/joycemara/european-fashion-store-multitable-dataset
  - **A clean dataset will be provided for the exercises! Don't download the raw one!**



Source: Amazon Prime Video

Exercises (KU)

- **Exercise 1: Data Modeling (25 points)**
  - Entity relationship (ER) modeling (15 points)
    - Resulting diagram needs to capture the main information of the dataset
  - Mapping ER diagrams into relational model (10 points)
    - Relationships, keys and data attributes need to match the dataset

Exercises (KU)

- **Exercise 2: Data Ingestion and Queries (30 points)**
  - Database and schema creation, and data ingestion (5 points)
    - Database should be complete
  - SQL query processing (25 points)
    - 5 SQL queries to answer questions on the data

Exercises (KU)

- **Exercise 3: APIs and Tuning (25 points)**
  - Database Connectors (10 points)
    - Connect via Python with the database and execute queries
  - Performance Testing an Index (15 points)
    - Use Python to create indices and improve query plan

Assessment

- Grading scheme (**> 40 points** needed for passing the course)
  - 0-40 points: 5
  - 41-50 points: 4
  - 51-60 points: 3
  - 61-70 points: 2
  - 71-80 points: 1

Course calendar

- **Part 1: Data Modeling and relational databases**
  - (1) Mon. 02.03.2026: Overview of databases
  - (2) Mon. 09.03.2026: Data modeling + **Exercise 1 presentation**
  - (3) Mon. 16.03.2026: Relational databases and normalization (+ bit of SQL for Ex. 1)
  - **No lecture on Mon. 23.03.2026 (time for Ex1 + VO exam #1)**
  - (4) Mon. 30.03.2026: Query languages (SQL) + **Exercise 2 presentation**
  - **Easter Break**
  - Mon. 20.04.2026: no lecture (time for Ex2)
  - (5) Mon. 27.04.2026: Relational algebra and query processing

Course calendar

- **Part 2: Tuning, APIs, NoSQL, Data Spaces**
  - No lecture on Mon. 04.05.2026 (Time for Ex2 + **VO exam #2**)
  - (6) Mon. 11.05.2026: Physical design and tuning / APIs and Python + **Exercise 3**
  - **No lecture on Mon. 18.05.2026** (time for Ex3)
  - Mon. 25.05.2026: Whit Monday **(Public holidays means no lecture)**
  - (7) Mon. 01.06.2026: NoSQL and Data Spaces.
  - Mon. 08.06.2026: no lecture (time to prepare for VO exam #3)
  - Mon. 15.06.2026: VO exam #3 **(means no lecture)**

Course calendar – Exercises and Exams Dates

- Exercise 1
  - Mon. 09.03.2026 – Tue. 31.03.2026
- Exercise 2
  - Wed. 01.04.2026 – Fri. 08.05.2026
- Exercise 3
  - Mon. 11.05.2026 – Tue. 02.06.2026
- **Individual exercises!**

Course calendar – Exercises and Exams Dates

- Exam 1
  - Mon. 23.03.2026
- Exam 2
  - Mon. 04.05.2026
- Exam 3
  - Mon. 15.06.2026

- **Choose one of these exams dates!**

Conclusions

- Summary
  - **Short introduction to databases**
    - **DB system = DBMS + DBs**
  - **Course organization**
    - **VO (3 exams per semester) → Lucas Iacono**
    - **KU (3 exercises) → Shafaq Siddiqi & Lucas Iacono**

- Next week
  - Data modeling + **Exercise 1**

- Questions / comments?
  - Discord